

Nadir Zouaoui

Electrical Engineer

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Oran, Algeria



Profile

State engineer specialized in renewable energy, with four years of experience in Oil & Gas and solar. I design, wire and commission electrical systems: skids, gas-pipeline SCADA, ATEX cabinets and PV plants, with over 1 MW of solar capacity delivered.

Professional experience

Electrical Engineer, RMBTECH EURL

Mar 2025 – Present

- **SCADA for Sonatrach gas pipelines.** Led the electrical engineering: site audit of 66 isolated stations (solar or grid powered) on GK3, which produced a detailed equipment condition status for each station (good, damaged, vandalized, etc.); deployment on GK1/GK2; pre-engineering of GR1/2/4/6 (integrator: SUPCON).
- **Critical equipment and ATEX.** Engineering for the replacement of an 8500 A rectifier for the AOA electrolysis system; design of an ATEX Ex "p" pressurized cabinet for Sonatrach GP2Z (Arzew).
- **Industrial rehabilitation.** Technical direction, redesign in Eplan, rewiring and automatic commissioning (Siemens S7-1200) of a BOSSAR packaging machine for GBFoods (Jumbo).
- **Agro-industry.** Electrical design for the SCADA system of the OAIC grain silo in Mesra.

Electrical Engineer, PROSKID EURL

Jul 2023 – Mar 2025

- Commissioned 27 Milton Roy skids and their solar installations.
- **ATEX Ex "d" flameproof enclosure.** Designed and built it for a skid, including component mounting and wiring.
- Design and sizing of skid electrical systems and off-grid solar power supplies; power supply, power and control diagrams.
- Design, assembly and wiring of electrical cabinets.

Photovoltaic Design Engineer, Sungy SPA

Oct 2022 – Jun 2023

- Over 300 kWp of installed solar capacity.
- 3D layout, sizing and simulation of PV systems; electrical sizing and diagrams for each installation.
- Commissioning and monitoring; energy sobriety plans; deployable mobile solar solutions.

Freelance

Sept 2018 – Present

Motion design, 3D/AR/VR simulation development and solar PV engineering. Portfolio on Behance, Freelancer, Fiverr.

Thesis project

5 kW autonomous photovoltaic microgrid

May – Jul 2021

- Sized and built a microgrid with a 3 kW PV array, an 800 Ah battery bank, a 50 kVA diesel generator and a three-input hybrid inverter (PV, battery, grid/genset), with electrical and synoptic diagrams.
- Installed the outdoor (PV array, supports) and indoor (inverter, batteries, generator, boxes, ATS) equipment, then commissioned and tested the system.

Education

State Engineer + Master II

2018 – 2021

École Nationale Polytechnique d'Oran – Maurice Audin (ENPO-MA). Valedictorian 2018–2021.

Preparatory cycle in sciences and technologies

2016 – 2018

ENPO-MA

Earlier experience

Document Controller, CEVIELEC SPA: ALRAR gas complex re-instrumentation	2022
Internship, SKTM: 12 MW Dhaya PV plant	2021
Internship, SDO Sonelgaz: MV-LV distribution network	2019
Internship, ENIE Solar: PV panel production	2019
Internship, CITAL: Sidi Bel Abbès tramway traction	2018

Skills and tools

Tools	EPLAN Electric P8 & Pro Panel, Archelios PV, AutoCAD Electrical, TIA Portal / Step 7, Microsoft Excel, Blender 3D, Adobe Suite
Domains	Oil & Gas, industrial equipment, hazardous areas (ATEX), MCC and switchgear, commissioning
Languages	Arabic (native), English (fluent), French (bilingual, DALF C1, 76/100)
Other	Driving license B, national service exempt